GATE 2017 Session 2 - Electronics and Communication Engineering

Answers and solutions for this paper were worked out by the compiler (no official key was supplied). Please verify against the official key.

GATE 2017 Session 2 | Q.01

Question Number : 1 **Correct : 1 Wrong : 0**

The rank of the matrix

$$\begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

is _____.

Answer: 4

Solution: The five rows sum to zero (incidence matrix of a 5-node cycle), so they are dependent; rank = 4.

GATE 2017 Session 2 | Q.02

Question Number : 2 **Correct : 1 Wrong : -0.33**

The general solution of the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 5y = 0$$

in terms of arbitrary constants K_1 and K_2 is

(A) $K_1 e^{(-1+\sqrt{6})x} + K_2 e^{(-1-\sqrt{6})x}$

(B) $K_1 e^{(-1+\sqrt{8})x} + K_2 e^{(-1-\sqrt{8})x}$

(C) $K_1 e^{(-2+\sqrt{6})x} + K_2 e^{(-2-\sqrt{6})x}$

(D) $K_1 e^{(-2+\sqrt{8})x} + K_2 e^{(-2-\sqrt{8})x}$

Answer: A

Solution: Characteristic equation lambda^2 + 2 lambda - 5 = 0 gives lambda = -1 +/- sqrt6.

GATE 2017 Session 2 | Q.03

Question Number : 3 **Correct : 1 Wrong : 0**

The smaller angle (in degrees) between the planes $x + y + z = 1$ and $2x - y + 2z = 0$ is _____.

Answer: 54.74

Solution: cos(theta) = n1.n2/(|n1||n2|) = (2 - 1 + 2)/(sqrt3 x 3) = 0.577, so theta = 54.74 degrees.

GATE 2017 Session 2 | Q.04

Question Number : 4 **Correct : 1 Wrong : -0.33**

The residues of a function

$$f(z) = \frac{1}{(z-4)(z+1)^3}$$

are

(A) $\frac{-1}{27}$ and $\frac{-1}{125}$

(B) $\frac{1}{125}$ and $\frac{-1}{125}$

(C) $\frac{-1}{27}$ and $\frac{1}{5}$

(D) $\frac{1}{125}$ and $\frac{-1}{5}$

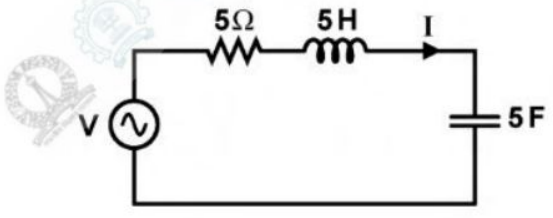
Answer: B

Solution: Residue at z = 4 is 1/5^3 = 1/125; at the triple pole z = -1 it is (1/2) d^2/dz^2 [1/(z-4)] at z = -1 = 1/(-5)^3 = -1/125.

GATE 2017 Session 2 | Q.05

Question Number : 5 **Correct : 1 Wrong : 0**

In the circuit shown, V is a sinusoidal voltage source. The current I is in phase with voltage V. The ratio $\frac{\text{amplitude of voltage across the capacitor}}{\text{amplitude of voltage across the resistor}}$ is _____.



Answer: 0.2

Solution: At resonance w = 1/sqrt(LC) = 0.2 rad/s; Xc = 1/(wC) = 1 ohm, R = 5 ohm: ratio = 1/5 = 0.2.

GATE 2017 Session 2 | Q.06

Question Number : 6 **Correct : 1 Wrong : 0**

A connection is made consisting of resistance A in series with a parallel combination of resistances B and C. Three resistors of value 10 Ω, 5 Ω, 2 Ω are provided. Consider all possible permutations of the given resistors into the positions A, B, C, and identify the configurations with maximum possible overall resistance, and also the ones with minimum possible overall resistance. The ratio of maximum to minimum values of the resistances (up to second decimal place) is _____.

Answer: 2.14

Solution: Maximum: A = 10 with 5||2 = 1.43 gives 11.43 ohm. Minimum: A = 2 with 10||5 = 3.33 gives 5.33 ohm. Ratio = 2.14.

GATE 2017 Session 2 | Q.07

Question Number : 7 **Correct : 1 Wrong : -0.33**

An LTI system with unit sample response $h[n] = 5\delta[n] - 7\delta[n-1] + 7\delta[n-3] - 5\delta[n-4]$ is a

(A) low-pass filter

(B) high-pass filter

(C) band-pass filter

(D) band-stop filter

Answer: C

Solution: H(z) has zeros at z = 1 and z = -1 (sum of coefficients and alternating sum are zero): band-pass filter.

GATE 2017 Session 2 | Q.08

Question Number : 8 **Correct : 1 Wrong : -0.33**

The input $x(t)$ and the output $y(t)$ of a continuous-time system are related as

$$y(t) = \int_{t-T}^{t} x(u)\,du.$$

The system is

(A) linear and time-variant

(B) linear and time-invariant

(C) non-linear and time-variant

(D) non-linear and time-invariant

Answer: B

Solution: The moving-window integral is linear and time invariant.

GATE 2017 Session 2 | Q.09

Question Number : 9 **Correct : 1 Wrong : -0.33**

An n-channel enhancement mode MOSFET is biased at $V_{GS} > V_{TH}$ and $V_{DS} > (V_{GS} - V_{TH})$, where V_{GS} is the gate-to-source voltage, V_{DS} is the drain-to-source voltage and V_{TH} is the threshold voltage. Considering channel length modulation effect to be significant, the MOSFET behaves as a

(A) voltage source with zero output impedance

(B) voltage source with non-zero output impedance

(C) current source with finite output impedance

(D) current source with infinite output impedance

Answer: C

Solution: With channel-length modulation the saturated MOSFET acts as a current source with finite output impedance.

GATE 2017 Session 2 | Q.10

Question Number : 10 **Correct : 1 Wrong : -0.33**

An npn bipolar junction transistor (BJT) is operating in the active region. If the reverse bias across the base-collector junction is increased, then

(A) the effective base width increases and common-emitter current gain increases

(B) the effective base width increases and common-emitter current gain decreases

(C) the effective base width decreases and common-emitter current gain increases

(D) the effective base width decreases and common-emitter current gain decreases

Answer: C

Solution: Higher reverse bias on the collector junction narrows the effective base, raising current gain (Early effect).

GATE 2017 Session 2 | Q.11

Question Number : 11 **Correct : 1 Wrong : 0**

Consider an n-channel MOSFET having width W, length L, electron mobility in the channel μ_n and oxide capacitance per unit area C_{ox}. If gate-to-source voltage $V_{GS} = 0.7$ V, drain-to-source voltage $V_{DS} = 0.1$ V, $(\mu_n C_{ox}) = 100$ μA/V^2, threshold voltage $V_{TH} = 0.3$ V and $(W/L) = 50$, then the transconductance g_m (in mA/V) is _____

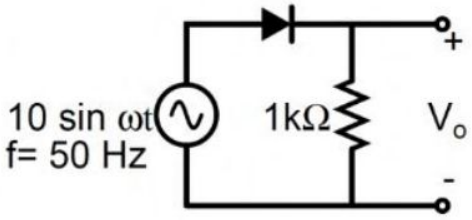
Answer: 0.5

Solution: VDS = 0.1 V < VGS - VTH = 0.4 V: triode region, gm = mu Cox (W/L) VDS = 100 uA x 50 x 0.1 = 0.5 mA/V.

GATE 2017 Session 2 | Q.12

Question Number : 12 **Correct : 1 Wrong : 0**

The output V_0 of the diode circuit shown in the figure is connected to an averaging DC voltmeter. The reading on the DC voltmeter in Volts, neglecting the voltage drop across the diode, is _____.

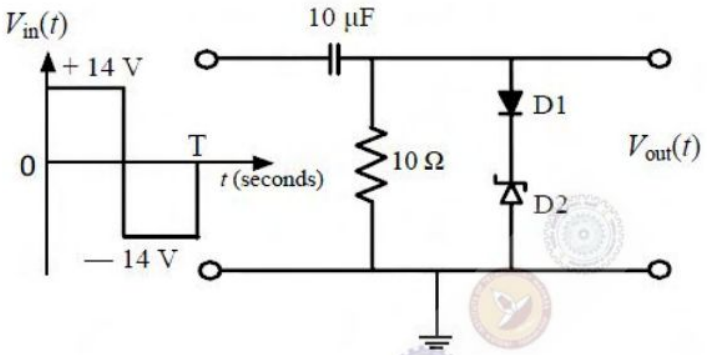


Answer: 3.18

Solution: Half-wave rectifier output average = Vm/pi = 10/pi = 3.18 V.

Question Number : 13 **Correct : 1 Wrong : -0.33**

In the figure, D1 is a real silicon *pn* junction diode with a drop of 0.7 V under forward bias condition and D2 is a Zener diode with breakdown voltage of − 6.8 V. The input $V_{in}(t)$ is a periodic square wave of period T, whose one period is shown in the figure.



Assuming $10\tau << T$, where τ is the time constant of the circuit, the maximum and minimum values of the output waveform are respectively,

(A) 7.5 V and −20.5 V

(B) 6.1 V and −21.9 V

(C) 7.5 V and −21.2 V

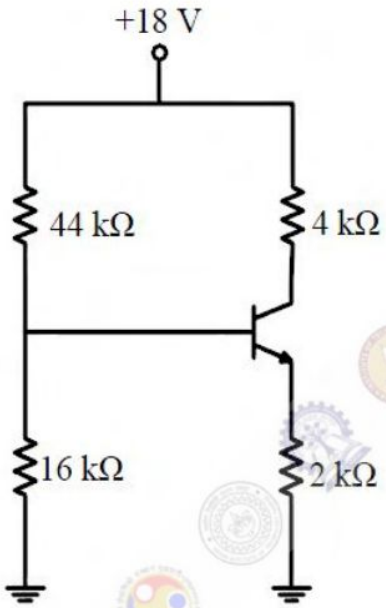
(D) 6.1 V and −22.6 V

Answer: A

Solution: The capacitor clamps the waveform; the Zener branch limits the positive peak to 6.8 + 0.7 = 7.5 V and the peak-to-peak 28 V gives a minimum of -20.5 V.

Question Number : 14 **Correct : 1 Wrong : 0**

Consider the circuit shown in the figure. Assume base-to-emitter voltage $V_{BE} = 0.8$ V and common-base current gain (α) of the transistor is unity.



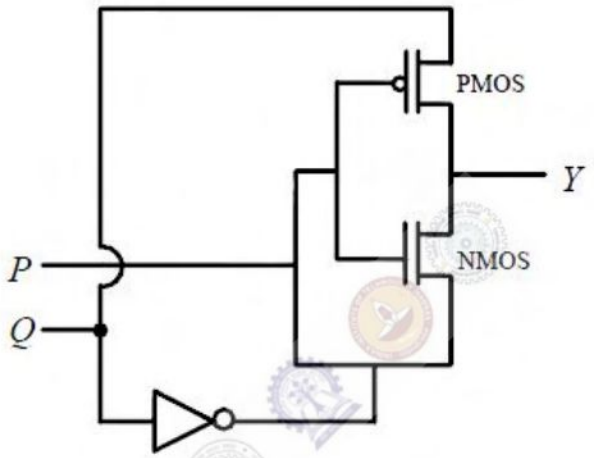
The value of the collector-to-emitter voltage V_{CE} (in volt) is _____

Answer: 6

Solution: VB = 18 x 16/60 = 4.8 V, VE = 4.0 V, IE = IC = 2 mA; VC = 18 - 4k x 2 mA = 10 V; VCE = 10 - 4 = 6 V.

Question Number : 15 **Correct : 1 Wrong : -0.33**

For the circuit shown in the figure, P and Q are the inputs and Y is the output.



The logic implemented by the circuit is

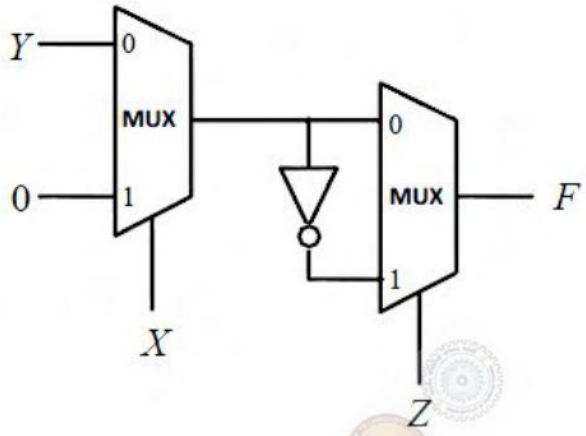
(A) XNOR (B) XOR (C) NOR (D) OR

Answer: A

Solution: (Low confidence) The complementary pass-transistor network with P and Q (and Q inverted) realises XNOR.

Question Number : 16 **Correct : 1 Wrong : -0.33**

Consider the circuit shown in the figure.



The Boolean expression F implemented by the circuit is

(A) $\bar{X}\,\bar{Y}\,\bar{Z} + X\,Y + \bar{Y}\,Z$

(B) $\bar{X}\,Y\,\bar{Z} + X\,Z + \bar{Y}\,Z$

(C) $\bar{X}\,Y\,\bar{Z} + X\,Y + \bar{Y}\,Z$

(D) $\bar{X}\,\bar{Y}\,\bar{Z} + X\,Z + \bar{Y}\,Z$

Answer: A

Solution: (Low confidence) Evaluating the two cascaded multiplexers gives option A.

GATE 2017 Session 2 | Q.17

Question Number : 17 **Correct : 1 Wrong : -0.33**

In a DRAM,

(A) periodic refreshing is not required

(B) information is stored in a capacitor

(C) information is stored in a latch

(D) both read and write operations can be performed simultaneously

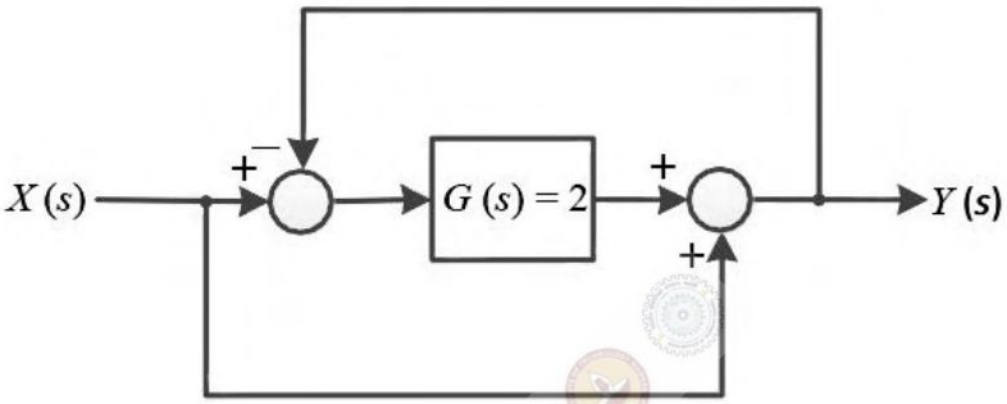
Answer: B

Solution: A DRAM cell stores information as charge on a capacitor and needs periodic refresh.

GATE 2017 Session 2 | Q.18

Question Number : 18 **Correct : 1 Wrong : 0**

For the system shown in the figure, $Y(s)/X(s) =$ _____



Answer: 1

Solution: Y = 2(X - Y) + Y gives Y = 2X - Y, so Y/X = 1.

GATE 2017 Session 2 | Q.19

Question Number : 19 **Correct : 1 Wrong : 0**

Consider the state space realization

$$\begin{bmatrix}\dot{x}_1(t)\\ \dot{x}_2(t)\end{bmatrix} = \begin{bmatrix}0 & 0\\ 0 & -9\end{bmatrix}\begin{bmatrix}x_1(t)\\ x_2(t)\end{bmatrix} + \begin{bmatrix}0\\ 45\end{bmatrix}u(t)\,,\ \text{with the initial condition } \begin{bmatrix}x_1(0)\\ x_2(0)\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix},$$

where $u(t)$ denotes the unit step function. The value of $\lim_{t\to\infty}\left|\sqrt{x_1^2(t)+x_2^2(t)}\right|$ is _____.

Answer: 5

Solution: x2' = -9 x2 + 45 u settles to x2 = 5; x1' = 0 stays at 0 from zero initial state; magnitude 5.

GATE 2017 Session 2 | Q.20

Question Number : 20 **Correct : 1 Wrong : -0.33**

Which of the following statements is **incorrect**?

(A) Lead compensator is used to reduce the settling time.

(B) Lag compensator is used to reduce the steady state error.

(C) Lead compensator may increase the order of a system.

(D) Lag compensator always stabilizes an unstable system.

Answer: D

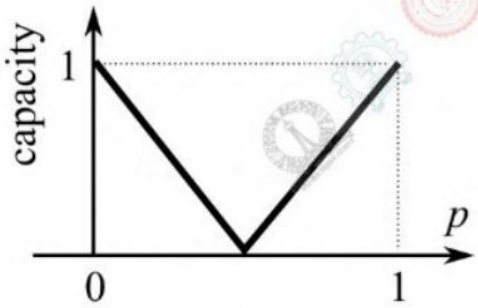
Solution: A lag compensator does not always stabilise an unstable system.

GATE 2017 Session 2 | Q.21

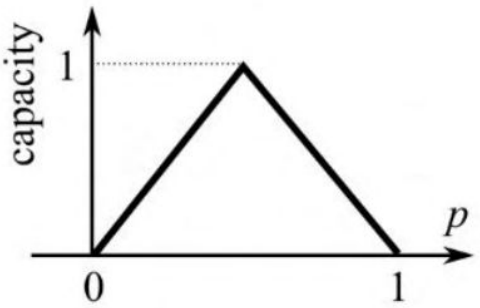
Question Number : 21 **Correct : 1 Wrong : -0.33**

Which one of the following graphs shows the Shannon capacity (channel capacity) in bits of a memoryless binary symmetric channel with crossover probability p?

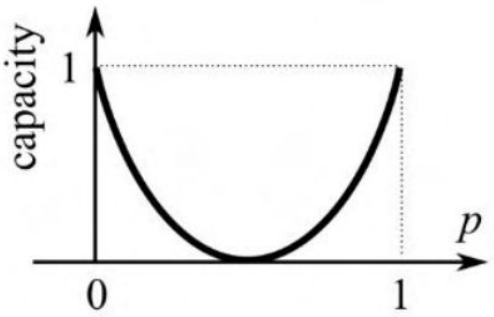
(A)



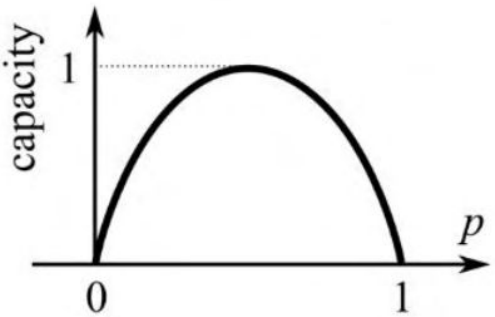
(B)



(C)



(D)



Answer: C

Solution: BSC capacity C = 1 - H(p) is 1 at p = 0 and p = 1 and 0 at p = 0.5 (U shape).

GATE 2017 Session 2 | Q.22

Question Number : 22 **Correct : 1 Wrong : 0**

Consider the random process

$$X(t) = U + Vt,$$

where U is a zero-mean Gaussian random variable and V is a random variable uniformly distributed between 0 and 2. Assume that U and V are statistically independent. The mean value of the random process at $t = 2$ is _____

Answer: 1

Solution: E[X(2)] = E[U] + 2E[V] = 0 + 2 x 0.5 = 1.

GATE 2017 Session 2 | Q.23

Question Number : 23 **Correct : 1 Wrong : 0**

A sinusoidal message signal is converted to a PCM signal using a uniform quantizer. The required signal-to-quantization noise ratio (SQNR) at the output of the quantizer is 40 dB. The minimum number of bits per sample needed to achieve the desired SQNR is _____

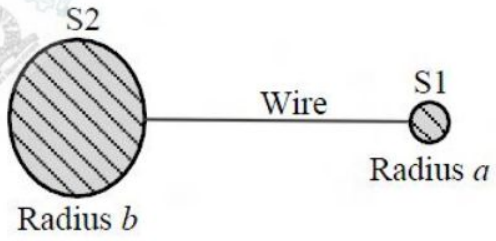
Answer: 7

Solution: SQNR = 1.76 + 6.02 n >= 40 gives n >= 6.35, so 7 bits.

GATE 2017 Session 2 | Q.24

Question Number : 24 **Correct : 1 Wrong : -0.33**

Two conducting spheres S1 and S2 of radii a and b ($b > a$) respectively, are placed far apart and connected by a long, thin conducting wire, as shown in the figure.



For some charge placed on this structure, the potential and surface electric field on S1 are V_a and E_a, and that on S2 are V_b and E_b, respectively. Then, which of the following is **CORRECT**?

(A) $V_a = V_b$ and $E_a < E_b$

(B) $V_a > V_b$ and $E_a > E_b$

(C) $V_a = V_b$ and $E_a > E_b$

(D) $V_a > V_b$ and $E_a = E_b$

Answer: C

Solution: The two spheres are at the same potential; the smaller sphere has the higher surface charge density and field: Va = Vb and Ea > Eb.

GATE 2017 Session 2 | Q.25

Question Number : 25 **Correct : 1 Wrong : 0**

A two-wire transmission line terminates in a television set. The VSWR measured on the line is 5.8. The percentage of power that is reflected from the television set is _____

Answer: 49.8

Solution: |Gamma| = (5.8-1)/(5.8+1) = 0.706; reflected power = 0.706^2 = 49.8 percent.

Question Number : 26 **Correct : 2 Wrong : -0.66**

The values of the integrals

$$\int_0^1 \left(\int_0^1 \frac{x-y}{(x+y)^3} dy \right) dx$$

and

$$\int_0^1 \left(\int_0^1 \frac{x-y}{(x+y)^3} dx \right) dy$$

are

(A) same and equal to 0.5

(B) same and equal to − 0.5

(C) 0.5 and − 0.5, respectively

(D) − 0.5 and 0.5, respectively

Answer: C

Solution: Inner integral of (x-y)/(x+y)^3 over y gives 1/(x+1)^2, whose integral is 0.5; with the order reversed the value is -0.5.

GATE 2017 Session 2 | Q.27

Question Number : 27 **Correct : 2 Wrong : -0.66**

An integral I over a counterclockwise circle C is given by

$$I = \oint_C \frac{z^2-1}{z^2+1} e^z \, dz.$$

If C is defined as $|z| = 3$, then the value of I is

(A) $-\pi i \sin(1)$ (B) $-2\pi i \sin(1)$ (C) $-3\pi i \sin(1)$ (D) $-4\pi i \sin(1)$

Answer: D

Solution: Residues of (z^2-1)e^z/(z^2+1) at +/- i sum to -2 sin(1); I = 2 pi i x (-2 sin 1) = -4 pi i sin(1).

GATE 2017 Session 2 | Q.28

Question Number : 28 **Correct : 2 Wrong : -0.66**

If the vector function $\vec{F} = \hat{a_x}(3y - k_1 z) + \hat{a_y}(k_2 x - 2z) - \hat{a_z}(k_3 y + z)$ is irrotational, then the values of the constants k_1, k_2 and k_3, respectively, are

(A) 0.3, −2.5, 0.5 (B) 0.0, 3.0, 2.0

(C) 0.3, 0.33, 0.5 (D) 4.0, 3.0, 2.0

Answer: D

Solution: Zero curl needs k3 = 2 (x-component) and k2 = 3 (z-component); k1 is free: (4, 3, 2).

GATE 2017 Session 2 | Q.29

Question Number : 29 **Correct : 2 Wrong : 0**

Passengers try repeatedly to get a seat reservation in any train running between two stations until they are successful. If there is 40% chance of getting reservation in any attempt by a passenger, then the average number of attempts that passengers need to make to get a seat reserved is

_____

Answer: 2.5

Solution: Geometric distribution with p = 0.4: mean 1/p = 2.5.

GATE 2017 Session 2 | Q.30

Question Number : 30 **Correct : 2 Wrong : 0**

The minimum value of the function $f(x) = \frac{1}{3}x(x^2 - 3)$ in the interval $-100 \leq x \leq 100$ occurs at $x =$ _____.

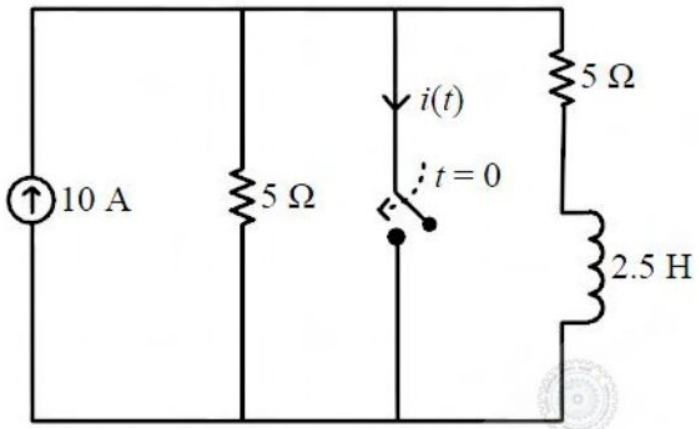
Answer: -100

Solution: f = x^3/3 - x is increasing for large |x|; its minimum on [-100,100] is at the left end x = -100.

GATE 2017 Session 2 | Q.31

Question Number : 31 **Correct : 2 Wrong : 0**

The switch in the circuit, shown in the figure, was open for a long time and is closed at $t = 0$.



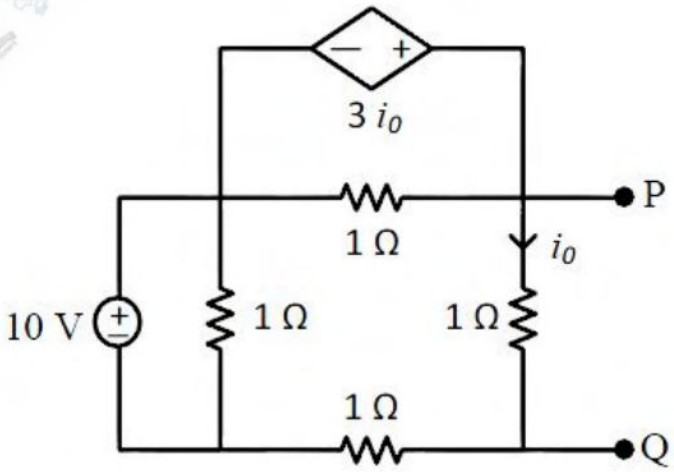
The current $i(t)$ (in ampere) at $t = 0.5$ seconds is _____

Answer: 8.16

Solution: Before closing the inductor carries 5 A; after closing the switch it decays as 5 e^{-2t}; i = 10 - 5 e^{-2t} = 8.16 A at 0.5 s.

Question Number : 32 **Correct : 2 Wrong : 0**

Consider the circuit shown in the figure.



The Thevenin equivalent resistance (in Ω) across P-Q is _____

Answer: 1.00

Solution: (Low confidence) Thevenin resistance with the dependent source evaluated by the test-source method.

GATE 2017 Session 2 | Q.33

Question Number : 33 **Correct : 2 Wrong : 0**

Consider an LTI system with magnitude response

$$|H(f)| = \begin{cases} 1 - \dfrac{|f|}{20}, & |f| \le 20 \\ 0, & |f| > 20 \end{cases}$$

and phase response

$$\arg\{H(f)\} = -2f.$$

If the input to the system is

$$x(t) = 8\cos\left(20\pi t + \frac{\pi}{4}\right) + 16\sin\left(40\pi t + \frac{\pi}{8}\right) + 24\cos\left(80\pi t + \frac{\pi}{16}\right),$$

then the average power of the output signal $y(t)$ is _____

Answer: 8

Solution: Only the 10 Hz component passes with gain 0.5 (amplitude 4); the 20 Hz and 40 Hz components are blocked. Power = 4^2/2 = 8.

GATE 2017 Session 2 | Q.34

Question Number : 34 **Correct : 2 Wrong : 0**

The transfer function of a causal LTI system is $H(s) = 1/s$. If the input to the system is $x(t) = [\sin(t)/\pi t]\, u(t)$, where $u(t)$ is a unit step function, the system output $y(t)$ as $t \to \infty$ is _____

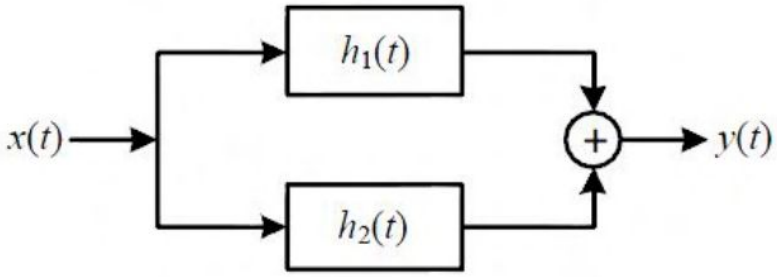
Answer: 0.5

Solution: y(inf) = integral from 0 to infinity of sin(t)/(pi t) dt = (1/pi)(pi/2) = 0.5.

GATE 2017 Session 2 | Q.35

Question Number : 35 **Correct : 2 Wrong : 0**

Consider the parallel combination of two LTI systems shown in the figure.



The impulse responses of the systems are

$$h_1(t) = 2\delta(t+2) - 3\delta(t+1)$$

$$h_2(t) = \delta(t-2).$$

If the input $x(t)$ is a unit step signal, then the energy of $y(t)$ is _____

Answer: 7

Solution: y(t) = 2u(t+2) - 3u(t+1) + u(t-2): value 2 on (-2,-1), -1 on (-1,2), 0 elsewhere. Energy = 4 x 1 + 1 x 3 = 7.

GATE 2017 Session 2 | Q.36

Question Number : 36 **Correct : 2 Wrong : 0**

A MOS capacitor is fabricated on p-type Si (Silicon) where the metal work function is 4.1 eV and electron affinity of Si is 4.0 eV. $E_C - E_F = 0.9$ eV, where E_C and E_F are the conduction band minimum and the Fermi energy levels of Si, respectively. Oxide $\epsilon_r = 3.9$, $\epsilon_o = 8.85 \times 10^{-14}$ F/cm, oxide thickness $t_{ox} = 0.1$ µm and electronic charge $q = 1.6 \times 10^{-19}$ C. If the measured flat band voltage of this capacitor is -1 V, then the magnitude of the fixed charge at the oxide-semiconductor interface, in nC/cm^2, is _____.

Answer: 6.9

Solution: phi_s = 4.0 + 0.9 = 4.9 eV, phi_ms = 4.1 - 4.9 = -0.8 V; Qf/Cox = phi_ms - VFB = 0.2 V; Cox = 3.45e-8 F/cm^2 gives Qf = 6.9 nC/cm^2.

GATE 2017 Session 2 | Q.37

Question Number : 37 **Correct : 2 Wrong : 0**

For a particular intensity of incident light on a silicon *pn* junction solar cell, the photocurrent density (J_L) is 2.5 mA/cm^2 and the open-circuit voltage (V_{oc}) is 0.451 V. Consider thermal voltage (V_T) to be 25 mV. If the intensity of the incident light is increased by 20 times, assuming that the temperature remains unchanged, V_{oc} (in volts) will be _____

Answer: 0.526

Solution: Voc increases by Vt ln 20 = 25 mV x 3.0 = 75 mV: 0.451 + 0.075 = 0.526 V.

GATE 2017 Session 2 | Q.38

Question Number : 38 **Correct : 2 Wrong : -0.66**

Two *n*-channel MOSFETs, T1 and T2, are identical in all respects except that the width of T2 is double that of T1. Both the transistors are biased in the saturation region of operation, but the gate overdrive voltage ($V_{GS} - V_{TH}$) of T2 is double that of T1, where V_{GS} and V_{TH} are the gate-to-source voltage and threshold voltage of the transistors, respectively. If the drain current and transconductance of T1 are I_{D1} and g_{m1} respectively, the corresponding values of these two parameters for T2 are

(A) $8I_{D1}$ and $2g_{m1}$ (B) $8I_{D1}$ and $4g_{m1}$ (C) $4I_{D1}$ and $4g_{m1}$ (D) $4I_{D1}$ and $2g_{m1}$

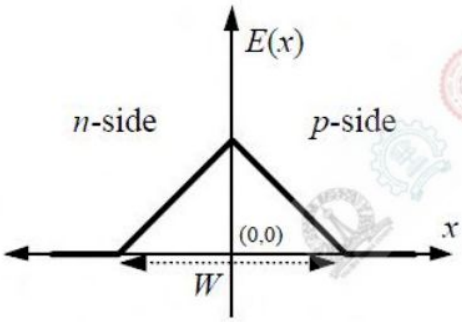
Answer: B

Solution: ID is proportional to W x Vov^2: 2 x 4 = 8 times; gm is proportional to W x Vov: 2 x 2 = 4 times.

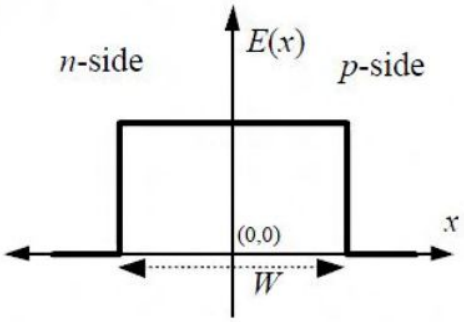
Question Number : 39 **Correct : 2 Wrong : -0.66**

An abrupt pn junction (located at $x = 0$) is uniformly doped on both p and n sides. The width of the depletion region is W and the electric field variation in the x-direction is $E(x)$. Which of the following figures represents the electric field profile near the pn junction?

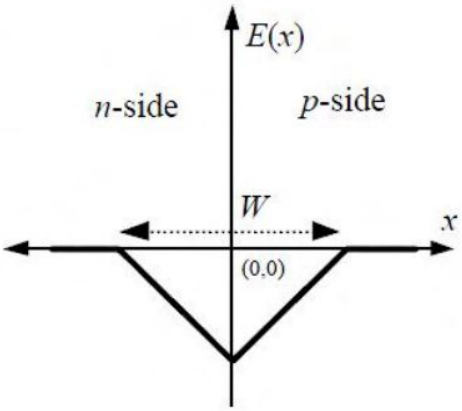
(A)



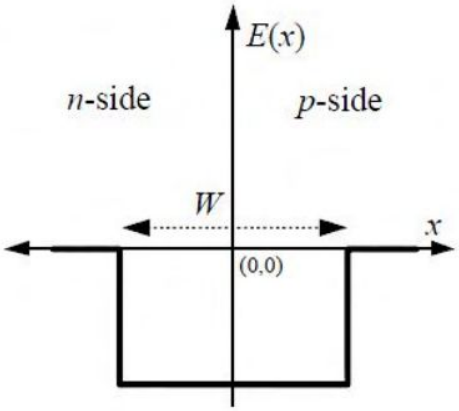
(B)



(C)



(D)

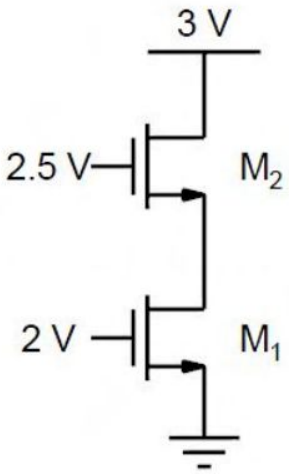


Answer: A

Solution: The field in a pn junction is a triangle peaking at the metallurgical junction, directed from n to p.

Question Number : 40 **Correct : 2 Wrong : -0.66**

Assuming that transistors M_1 and M_2 are identical and have a threshold voltage of 1 V, the state of transistors M_1 and M_2 are respectively



(A) Saturation, Saturation

(B) Linear, Linear

(C) Linear, Saturation

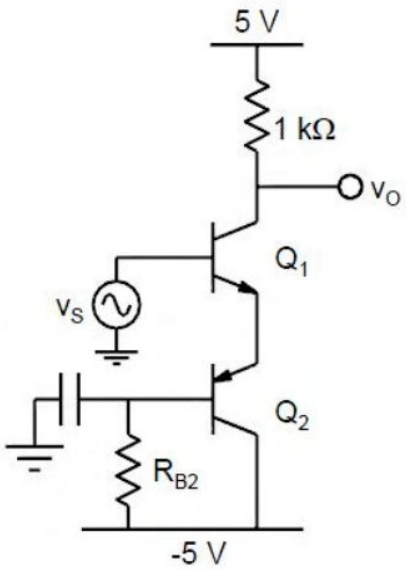
(D) Saturation, Linear

Answer: C

Solution: M2 is always saturated; solving the current equality gives V_mid = 0.59 V < 1 V, so M1 is in the linear region.

Question Number : 41 **Correct : 2 Wrong : 0**

In the circuit shown, transistors Q_1 and Q_2 are biased at a collector current of 2.6 mA. Assuming that transistor current gains are sufficiently large to assume collector current equal to emitter current and thermal voltage of 26 mV, the magnitude of voltage gain V_o/V_s in the mid-band frequency range is _____ (up to second decimal place).

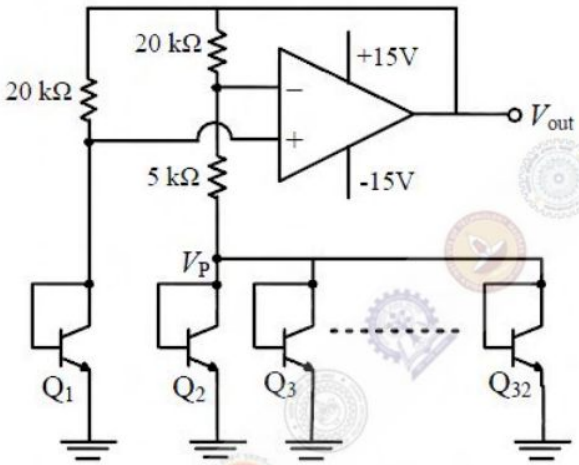


Answer: 100

Solution: Cascode (common-collector/common-base arrangement): gain = gm RC = (2.6 mA/26 mV)(1k) = 100.

Question Number : 42 **Correct : 2 Wrong : 0**

In the voltage reference circuit shown in the figure, the op-amp is ideal and the transistors Q_1, $Q_2, \ldots, Q_{32}$ are identical in all respects and have infinitely large values of common-emitter current gain (β). The collector current (I_c) of the transistors is related to their base-emitter voltage (V_{BE}) by the relation $I_C = I_S \exp(V_{BE}/V_T)$, where I_S is the saturation current. Assume that the voltage V_p shown in the figure is 0.7 V and the thermal voltage $V_T = 26$ mV.



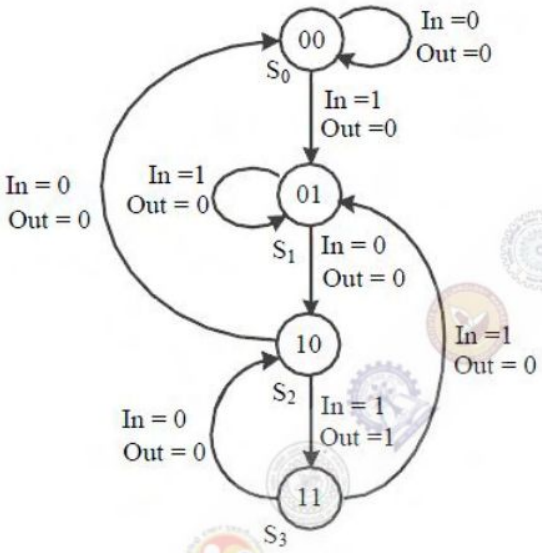
The output voltage V_{out} (in volts) is _____

Answer: 1.06

Solution: VBE(32 transistors) = 0.7 - 26 mV ln 32 = 0.61 V; I = (0.7 - 0.61)/5k = 18 uA; Vout = 0.7 + 20k x 18 uA = 1.06 V.

Question Number : 43 **Correct : 2 Wrong : 0**

The state diagram of a finite state machine (FSM) designed to detect an overlapping sequence of three bits is shown in the figure. The FSM has an input 'In' and an output 'Out'. The initial state of the FSM is S_0.



If the input sequence is 10101101001101, starting with the left-most bit, then the number of times 'Out' will be 1 is _____

Answer: 4

Solution: The FSM detects the overlapping pattern 101; in 10101101001101 it occurs 4 times.

Question Number : 44 **Correct : 2 Wrong : 0**

Figure I shows a 4-bit ripple carry adder realized using full adders and Figure II shows the circuit of a full-adder (FA). The propagation delay of the XOR, AND and OR gates in Figure II are 20 ns, 15 ns and 10 ns, respectively. Assume all the inputs to the 4-bit adder are initially reset to 0.

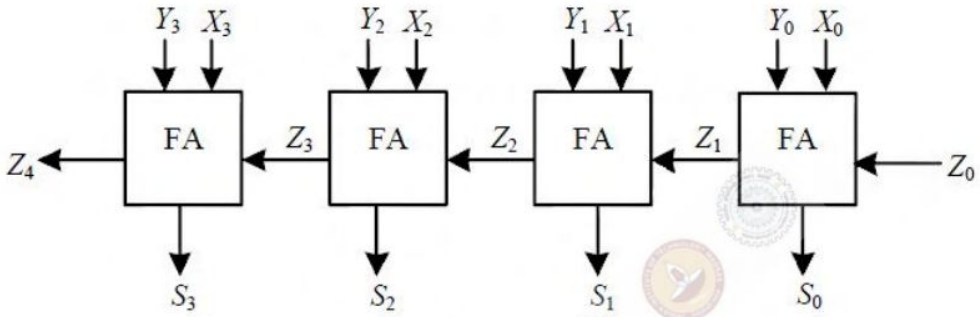


Figure I

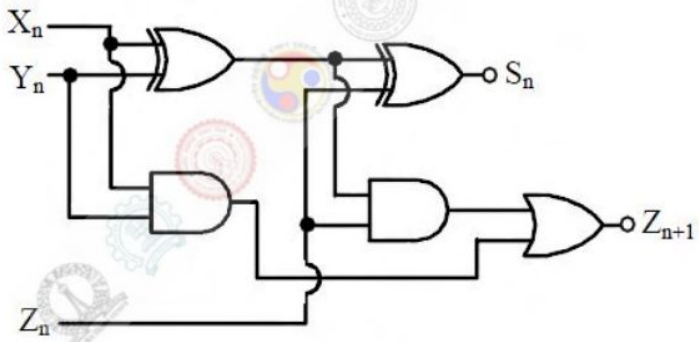


Figure II

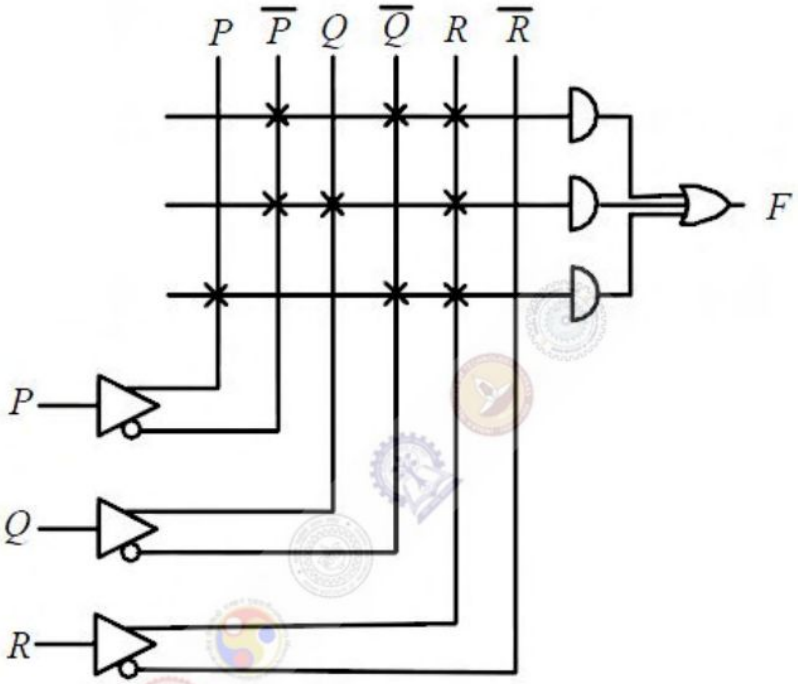
At $t = 0$, the inputs to the 4-bit adder are changed to $X_3X_2X_1X_0 = 1100$, $Y_3Y_2Y_1Y_0 = 0100$ and $Z_0 = 1$. The output of the ripple carry adder will be stable at t (in ns) = _____

Answer: 120

Solution: Carry delay per stage 15 + 10 = 25 ns after the first stage (45 ns); Z4 is stable at 45 + 3 x 25 = 120 ns.

Question Number : 45 **Correct : 2 Wrong : -0.66**

A programmable logic array (PLA) is shown in the figure.



The Boolean function F implemented is

(A) $\bar{P}\,\bar{Q}\,R + \bar{P}\,Q\,R + P\,\bar{Q}\,\bar{R}$

(B) $(\bar{P} + \bar{Q} + R)\,(\bar{P} + Q + R)\,(P + \bar{Q} + \bar{R})$

(C) $\bar{P}\,\bar{Q}\,R + \bar{P}\,Q\,R + P\,\bar{Q}\,R$

(D) $(\bar{P} + \bar{Q} + R)\,(\bar{P} + Q + R)\,(P + \bar{Q} + R)$

Answer: C

Solution: (Low confidence) The three product terms read from the AND plane are P'Q'R, P'QR and PQ'R.

GATE 2017 Session 2 | Q.46

Question Number : 46 **Correct : 2 Wrong : 0**

A unity feedback control system is characterized by the open-loop transfer function

$$G(s) = \frac{2(s+1)}{s^3 + ks^2 + 2s + 1}$$

The value of k for which the system oscillates at 2 rad/s is _____

Answer: 0.75

Solution: Characteristic equation s^3 + k s^2 + 4 s + 3 = 0; sustained oscillation at 2 rad/s needs 4k = 3, so k = 0.75.

GATE 2017 Session 2 | Q.47

Question Number : 47 **Correct : 2 Wrong : -0.66**

A second-order LTI system is described by the following state equations.

$$\frac{d}{dt}x_1(t) - x_2(t) = 0$$

$$\frac{d}{dt}x_2(t) + 2x_1(t) + 3x_2(t) = r(t)$$

where $x_1(t)$ and $x_2(t)$ are the two state variables and $r(t)$ denotes the input. The output $c(t) = x_1(t)$. The system is

(A) undamped (oscillatory)

(B) underdamped

(C) critically damped

(D) overdamped

Answer: D

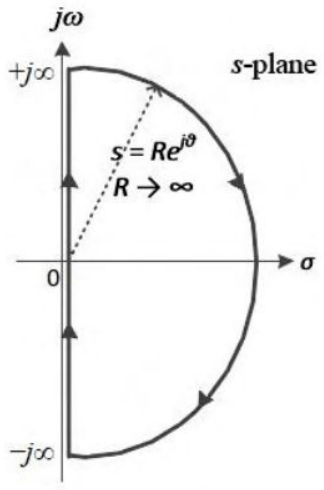
Solution: s^2 + 3s + 2 = 0: damping ratio 3/(2 sqrt2) = 1.06 > 1, overdamped.

Question Number : 48 **Correct : 2 Wrong : -0.66**

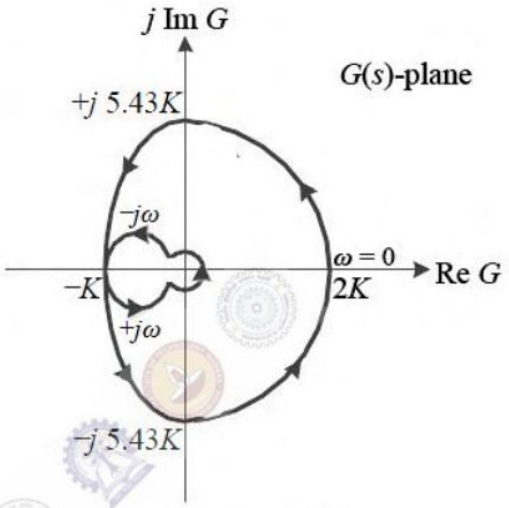
A unity feedback control system is characterized by the open-loop transfer function

$$G(s) = \frac{10K(s+2)}{s^3 + 3s^2 + 10}$$

The Nyquist path and the corresponding Nyquist plot of $G(s)$ are shown in the figures below.



Nyquist Path for $G(s)$



Nyquist Plot of $G(s)$

If $0 < K < 1$, then the number of poles of the closed-loop transfer function that lie in the right-half of the s-plane is

(A) 0 (B) 1 (C) 2 (D) 3

Answer: C

Solution: (Low confidence) With two open-loop RHP poles and the Nyquist plot for 0 < K < 1, two closed-loop poles remain in the RHP.

GATE 2017 Session 2 | Q.49

Question Number : 49 **Correct : 2 Wrong : -0.66**

The signal $x(t) = \sin(14000\pi t)$, where t is in seconds, is sampled at a rate of 9000 samples per second. The sampled signal is the input to an ideal lowpass filter with frequency response $H(f)$ as follows:

$$H(f) = \begin{cases} 1, & |f| \leq 12 \text{ kHz} \\ 0, & |f| > 12 \text{ kHz}. \end{cases}$$

What is the number of sinusoids in the output and their frequencies in kHz?

(A) Number = 1, frequency = 7

(B) Number = 3, frequencies = 2, 7, 11

(C) Number = 2, frequencies = 2, 7

(D) Number = 2, frequencies = 7, 11

Answer: B

Solution: 7 kHz sampled at 9 kHz has images at 2, 7, 11 and 16 kHz; the 12 kHz low-pass keeps 2, 7 and 11 kHz.

GATE 2017 Session 2 | Q.50

Question Number : 50 **Correct : 2 Wrong : 0**

The unmodulated carrier power in an AM transmitter is 5 kW. This carrier is modulated by a sinusoidal modulating signal. The maximum percentage of modulation is 50%. If it is reduced to 40%, then the maximum unmodulated carrier power (in kW) that can be used without overloading the transmitter is _____

Answer: 5.21

Solution: Total allowed power = 5(1 + 0.5^2/2) = 5.625 kW; with 40 percent modulation Pc(1 + 0.08) = 5.625 gives Pc = 5.21 kW.

GATE 2017 Session 2 | Q.51

Question Number : 51 **Correct : 2 Wrong : 0**

A modulating signal given by $x(t) = 5\sin(4\pi 10^3 t - 10\pi \cos 2\pi 10^3 t)$ V is fed to a phase modulator with phase deviation constant $k_p = 5$ rad/V. If the carrier frequency is 20 kHz, the instantaneous frequency (in kHz) at $t = 0.5$ ms is _____

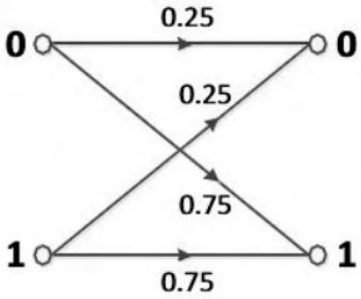
Answer: 70

Solution: dx/dt at 0.5 ms equals 20 pi x 10^3; frequency deviation = (kp/2 pi) dx/dt = 50 kHz; total 20 + 50 = 70 kHz.

GATE 2017 Session 2 | Q.52

Question Number : 52 **Correct : 2 Wrong : -0.66**

Consider a binary memoryless channel characterized by the transition probability diagram shown in the figure.



The channel is

(A) lossless

(B) noiseless

(C) useless

(D) deterministic

Answer: C

Solution: The output is independent of the input (equal conditional probabilities 0.25/0.75), so the channel is useless.

GATE 2017 Session 2 | Q.53

Question Number : 53 **Correct : 2 Wrong : 0**

An electron (q_1) is moving in free space with velocity 10^5 m/s towards a stationary electron (q_2) far away. The closest distance that this moving electron gets to the stationary electron before the repulsive force diverts its path is _____ $\times 10^{-8}$ m.

[Given, mass of electron $m = 9.11\times10^{-31}$ kg, charge of electron $e = -1.6\times10^{-19}$ C, and permittivity $\varepsilon_0 = (1/36\pi)\times10^{-9}$ F/m]

Answer: 5.06

Solution: At closest approach all kinetic energy becomes potential energy: (1/2) m v^2 = e^2/(4 pi eps0 d) gives d = 5.06e-8 m.

GATE 2017 Session 2 | Q.54

Question Number : 54 **Correct : 2 Wrong : -0.66**

Standard air-filled rectangular waveguides of dimensions $a = 2.29$ cm and $b = 1.02$ cm are designed for radar applications. It is desired that these waveguides operate only in the dominant TE_{10} mode with the operating frequency at least 25% above the cutoff frequency of the TE_{10} mode but not higher than 95% of the next higher cutoff frequency. The range of the allowable operating frequency f is

(A) $8.19\ \text{GHz} \le f \le 13.1\ \text{GHz}$

(B) $8.19\ \text{GHz} \le f \le 12.45\ \text{GHz}$

(C) $6.55\ \text{GHz} \le f \le 13.1\ \text{GHz}$

(D) $1.64\ \text{GHz} \le f \le 10.24\ \text{GHz}$

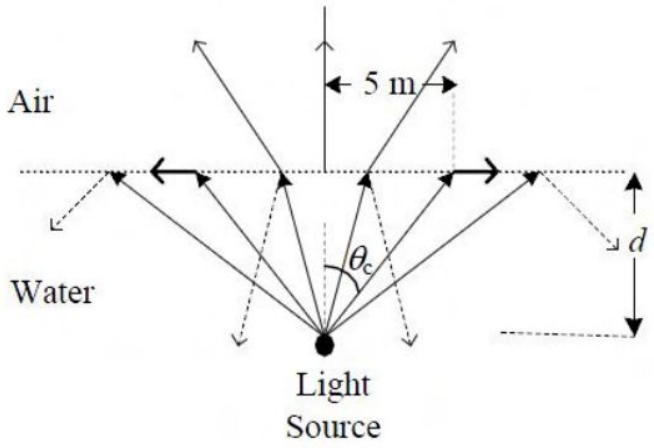
Answer: B

Solution: fc10 = 6.55 GHz and the next mode (TE20) cuts off at 13.1 GHz; 1.25 x 6.55 = 8.19 GHz and 0.95 x 13.1 = 12.45 GHz.

GATE 2017 Session 2 | Q.55

Question Number : 55 **Correct : 2 Wrong : 0**

The permittivity of water at optical frequencies is $1.75\varepsilon_0$. It is found that an isotropic light source at a distance d under water forms an illuminated circular area of radius 5 m, as shown in the figure. The critical angle is θ_c.



The value of d (in meter) is _____

Answer: 4.33

Solution: sin(theta_c) = 1/sqrt(1.75) so theta_c = 49.1 degrees; d = 5/tan(theta_c) = 4.33 m.

GATE 2017 Session 2 | Q.56

Question Number : 56 **Correct : 1 Wrong : -0.33**

The ninth and the tenth of this month are Monday and Tuesday _____.

(A) figuratively (B) retrospectively (C) respectively (D) rightfully

Answer: C

Solution: 'Respectively' matches Monday and Tuesday to the ninth and tenth.

GATE 2017 Session 2 | Q.57

Question Number : 57 **Correct : 1 Wrong : -0.33**

It is _____ to read this year's textbook _____the last year's.

(A) easier, than (B) most easy, than (C) easier, from (D) easiest, from

Answer: A

Solution: 'It is easier to read this year's textbook than the last year's.'

GATE 2017 Session 2 | Q.58

Question Number : 58 **Correct : 1 Wrong : -0.33**

A rule states that in order to drink beer, one must be over 18 years old. In a bar, there are 4 people. P is 16 years old, Q is 25 years old, R is drinking milkshake and S is drinking a beer. What must be checked to ensure that the rule is being followed?

(A) Only P's drink
(B) Only P's drink and S's age
(C) Only S's age
(D) Only P's drink, Q's drink and S's age

Answer: B

Solution: To enforce the rule check P's drink (must not be beer) and S's age (S drinks beer).

GATE 2017 Session 2 | Q.59

Question Number : 59 **Correct : 1 Wrong : -0.33**

Fatima starts from point P, goes North for 3 km, and then East for 4 km to reach point Q. She then turns to face point P and goes 15 km in that direction. She then goes North for 6 km. How far is she from point P, and in which direction should she go to reach point P?

(A) 8 km, East (B) 12 km, North (C) 6 km, East (D) 10 km, North

Answer: A

Solution: Q is at (4,3); walking 15 km toward P gives (-8,-6); then 6 km north gives (-8,0). She must go 8 km East to reach P.

GATE 2017 Session 2 | Q.60

Question Number : 60 **Correct : 1 Wrong : -0.33**

500 students are taking one or more courses out of Chemistry, Physics, and Mathematics. Registration records indicate course enrolment as follows: Chemistry (329), Physics (186), Mathematics (295), Chemistry and Physics (83), Chemistry and Mathematics (217), and Physics and Mathematics (63). How many students are taking all 3 subjects?

(A) 37 (B) 43 (C) 47 (D) 53

Answer: D

Solution: 500 = 810 - 363 + x gives x = 53.

GATE 2017 Session 2 | Q.61

Question Number : 61 **Correct : 2 Wrong : -0.66**

"If you are looking for a history of India, or for an account of the rise and fall of the British Raj, or for the reason of the cleaving of the subcontinent into two mutually antagonistic parts and the effects this mutilation will have in the respective sections, and ultimately on Asia, you will not find it in these pages; for though I have spent a lifetime in the country, I lived too near the seat of events, and was too intimately associated with the actors, to get the perspective needed for the impartial recording of these matters."

Which of the following statements best reflects the author's opinion?

(A) An intimate association does not allow for the necessary perspective.
(B) Matters are recorded with an impartial perspective.
(C) An intimate association offers an impartial perspective.
(D) Actors are typically associated with the impartial recording of matters.

Answer: A

Solution: The author says that being too near the events and too intimately associated with the actors prevents the perspective needed.

GATE 2017 Session 2 | Q.62

Question Number : 62 **Correct : 2 Wrong : -0.66**

Each of P, Q, R, S, W, X, Y and Z has been married at most once. X and Y are married and have two children P and Q. Z is the grandfather of the daughter S of P. Further, Z and W are married and are parents of R. Which one of the following must necessarily be FALSE?

(A) X is the mother-in-law of R
(B) P and R are not married to each other
(C) P is a son of X and Y
(D) Q cannot be married to R

Answer: B

Solution: S's other parent is a child of Z and W, i.e. R; so P and R must be married and statement B is false.

GATE 2017 Session 2 | Q.63

Question Number : 63 **Correct : 2 Wrong : -0.66**

1200 men and 500 women can build a bridge in 2 weeks. 900 men and 250 women will take 3 weeks to build the same bridge. How many men will be needed to build the bridge in one week?

(A) 3000 (B) 3300 (C) 3600 (D) 3900

Answer: C

Solution: 2(1200 m + 500 w) = 3(900 m + 250 w) gives w = 1.2 m; total work 3600 man-weeks, so 3600 men for one week.

GATE 2017 Session 2 | Q.64

Question Number : 64 **Correct : 2 Wrong : -0.66**

The number of 3-digit numbers such that the digit 1 is never to the immediate right of 2 is

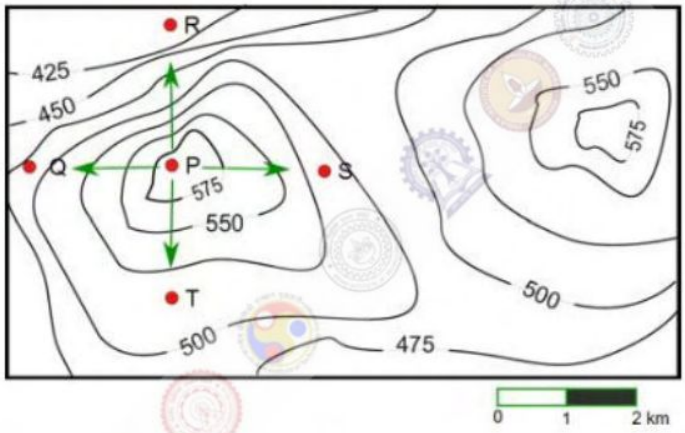
(A) 781 (B) 791 (C) 881 (D) 891

Answer: C

Solution: Three-digit numbers: 900 total; those containing '21' are 19; 900 - 19 = 881.

Question Number : 65 **Correct : 2 Wrong : -0.66**

A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot.



Which of the following is the steepest path leaving from P?

(A) P to Q (B) P to R (C) P to S (D) P to T

Answer: A

Solution: (Low confidence) The contour lines are closest together from P towards Q.